

UNIT – 1

1. Define Ohm's law.

→ Ohm's law states that current is directly proportional to voltage when temperature is constant.

$$V = IR$$

2. Define Kirchhoff's Current Law and Kirchhoff's Voltage Law.

→ KCL: Algebraic sum of currents at a node is zero.

KVL: Algebraic sum of voltages in a closed loop is zero.

3. Define Resistance, Inductance, and Capacitance and write their units.

→ Resistance: Opposes current flow, unit – Ohm (Ω).

Inductance: Opposes change in current, unit – Henry (H).

Capacitance: Stores electric charge, unit – Farad (F).

4. List the active and passive elements of an electrical circuit.

→ Active: Voltage source, current source.

Passive: Resistor, inductor, capacitor.

5. Find the current flowing through a 100 Ω resistor when 220 V is applied to it.

$$\rightarrow I = V / R = 220 / 100 = 2.2 \text{ A}$$

6. Give the statement of Thevenin's theorem and Norton's theorem.

→ Thevenin: Any linear network can be replaced by a single voltage source and series resistance.

Norton: Any linear network can be replaced by a current source and parallel resistance.

7. The voltage across a 5 ohm resistor is 10 volts. Find the current and power dissipated.

$$\rightarrow I = V / R = 10 / 5 = 2 \text{ A}$$

$$\text{Power} = VI = 10 \times 2 = 20 \text{ W}$$

8. Define time constant of RL and RC circuits.

→ Time constant is the time taken by current or voltage to reach 63.2% of its final value after switching.

For RL circuit: $\tau = L / R$

For RC circuit: $\tau = R C$

9. List various types of electrical sources used in electrical circuits.

→ Voltage source and current source.

AC source and DC source.

10. Give the statement of Superposition theorem.

→ In a linear circuit, total response is the sum of responses due to individual sources acting alone.

UNIT – 2

11. Define impedance of an AC circuit.

→ Impedance is total opposition to AC, including resistance and reactance.

Unit: Ohm (Ω)

12. State admittance and write its unit.

→ Admittance is reciprocal of impedance.

Unit: Siemens (S)

13. Define inductive reactance.

→ Inductive reactance is the opposition offered by an inductor to alternating current.

$$X_L = 2\pi f L$$

Unit: Ohm (Ω)

14. Define capacitive reactance.

→ Capacitive reactance is the opposition offered by a capacitor to alternating current.

$$X_C = 1 / (2\pi f C)$$

Unit: Ohm (Ω)

15. Define average value and RMS value of an AC quantity.

→ Average value is the mean value of an AC quantity over one half cycle.

Average value = $(2 / \pi) V_m$
RMS value is the effective value of AC.
RMS value = $V_m / \sqrt{2}$

16. Define form factor and peak factor.

→ Form factor = RMS / Average value

Peak factor = Maximum / RMS value

17. Define instantaneous value, frequency, and time period.

→ Instantaneous value: Value at a given instant.

Frequency: Number of cycles per second (Hz).

Time period: $T = 1/f$

18. Define phase and phase difference.

→ Phase: Position of waveform at an instant.

Phase difference: Angular difference between two waveforms.

19. Give the condition for series resonance and parallel resonance.

→ Series resonance: $X_L = X_C$

Parallel resonance: Inductive current = Capacitive current

20. Define resonant frequency and Q-factor.

→ Resonant frequency: Frequency at which resonance occurs.

Q-factor: Measure of sharpness of resonance.

UNIT – 3 (Transformers)

21. Give some conditions of an ideal transformer.

→ An ideal transformer has no losses in the core and windings.

It has zero winding resistance, no leakage flux, and infinite magnetic permeability of the core.

Hence, efficiency of an ideal transformer is 100%.

22. What is the turns ratio of a transformer?

→ Turns ratio of a transformer is the ratio of number of turns in the secondary winding to the number of turns in the primary winding.

It is equal to the ratio of secondary voltage to primary voltage.

Turns ratio = $N_2 / N_1 = V_2 / V_1$

23. List various losses in a transformer.

→ The losses in a transformer are mainly classified into copper losses and core losses.

Copper losses occur due to resistance of windings, while core losses include hysteresis loss and eddy current loss.

24. What is a step-up transformer?

→ A step-up transformer is a transformer in which the secondary voltage is higher than the primary voltage.

In this transformer, the number of turns in the secondary winding is greater than that in the primary winding.

25. List the applications and advantages of autotransformer.

→ Applications: Autotransformers are used in voltage control, motor starting, and power transmission systems.

Advantages: They are smaller in size, cost-effective, and have higher efficiency compared to two-winding transformers.

UNIT – 4 (Electrical Machines)

26. What type of material is used for brushes of a D.C machine?

→ Carbon or graphite is used for brushes in a DC machine.

These materials provide good conductivity and smooth contact with the commutator, reducing sparking and wear.

27. List few applications of DC motor.

→ DC motors are used where good speed control is required.

Examples include electric trains, cranes, hoists, elevators, rolling mills, and printing machines.

28. List few applications of induction motor.

→ Induction motors are widely used due to simple construction and low maintenance.

Examples include fans, pumps, compressors, air conditioners, and washing machines.

29. Calculate the speed of a 4-pole machine at a frequency of 50 Hz.

→ Synchronous speed is given by:

$$N_s = (120 \times f) / P$$

$$N_s = (120 \times 50) / 4$$

$$N_s = 1500 \text{ rpm}$$

30. Write the applications of synchronous generator.

→ Synchronous generators are mainly used in power stations for generating AC electrical power.

They are used in thermal, hydro, and nuclear power plants.

UNIT – 5 (Electrical Wiring & Safety)

31. List different types of cables.

→ The commonly used types of cables are single core, twin core, and multi-core cables.

They are used for domestic and industrial wiring applications.

32. What is the energy consumed by a 1000 W heater in 3 hours?

→ Electrical energy consumed is given by power multiplied by time.

$$\text{Energy} = \text{Power} \times \text{Time}$$

$$\text{Energy} = 1000 \text{ W} \times 3 \text{ h} = 3000 \text{ Wh} = 3 \text{ kWh}$$

33. What is the use of MCB?

→ MCB (Miniature Circuit Breaker) is used to protect electrical circuits from overload and short circuit conditions.

It automatically disconnects the supply during faults.

34. Define earthing and explain its importance.

→ Earthing is the process of connecting the electrical system to the ground.

It protects people and equipment by safely diverting fault current to the earth.

35. What is the full form of M.C.C.B?

→ The full form of MCCB is Moulded Case Circuit Breaker.